UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.
Instruction
Opcode
Address A
Memory
Operand
Figure 2
Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)
List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

Source: CSC429 (1).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A2 (5 marks)

Topic: Parallel and multiprocessor architecture

Complete the following taxonomy of parallel processor architecture with appropriate example.
Single Instruction Multiple
Data Stream
ii
Processor Organization
Multiple Instruction Multiple
Data Stream
v

Source: CSC429 (5).pdf, page(s) 6

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A3 (5 marks)

Topic: Data representation and number systems

Convert -6.75 to IEEE 754 single-precision format. Give the answer in hexadecimal and show the sign, exponent and mantissa fields.

Answer: -6.7510 = 0xC0D80000 (sign=1, exponent=10000001, fraction=10110000000000000000000)

- *1 mark for the sign bit.*
- *1 mark for the biased exponent.*
- *1 mark for the 23-bit fraction.*
- *1 mark for the correct 32-bit grouping.*
- *1 mark for the final hexadecimal value.*

QUESTION A4 (5 marks)

Topic: Computer generations and history

a) Discuss TWO (2) major technologies in the fourth computer generation and their impact on the modern computer.
b) Give ONE (1) example of using artificial intelligence (A1) in everyday life.

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

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QUESTION A5 (10 marks)

Topic: Computer architecture and CPU organisation

Given that the size of the data bus is 32 bits, the size of the address bus is 8 bits, and the size of the control bus is 4 bits.

a) Determine the size of the word and provide relevant justification.
b) Determine the size of the register and provide relevant justification.
c) Discuss what will happen if system interconnection is blocked from the perspective of the machine cycle.
d) Elaborate the role of CPU interconnection.

Source: CSC429 (17).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Computer architecture and CPU organisation

a)
State TWO (2) different types of bus.
b)
The size of a word is 32 bits and the size of the block is 256 bits. Determine of the following:
i)
The byte for word
ii)
The byte for block,
iii)
Size of the register,
iv)
Size of the data bus.

Source: CSC429 (13).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A7 (10 marks)

Topic: Input/output and interrupts

a) Discuss THREE (3) types of interfaces to extemal devices.
b) Classify the following input/output devices to their respective 1/0 module:
i)
Network interface card
ii)
Scanner
Touch Screen
iv)
Register

Source: CSC429 (7).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Input/output and interrupts

a) Explain the problem in polling 1/0 method.
b) Explain how buffering can help to improve the device overall performance?
c) Discuss how DMA technique can help the 1/0 transfer performance instead of using polling method and interrupt 1/0 method?

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

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QUESTION A9 (10 marks)

Topic: Memory hierarchy and cache

a) Explain how the bits are stored in the SRAM and DFiAM.
b) Why is SRAM technology faster than DRAM technology?
c) Compare SRAM and DRAM in terms of cost and power consumption.

Source: CSC429 (20).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Data representation and number systems

Answer both parts. Show your complete workflow.
a) Convert 137710 to hexadecimal.
b) Convert 4B816 to decimal.

a) Convert 137710 to hexadecimal.

Answer: 137710 = 56116

- *1 mark for repeated division by 16.*
- *1 mark for correct remainders.*
- *1 mark for reading remainders in reverse order.*
- *1 mark for the correct hexadecimal digits.*
- *1 mark for the base subscript.*

b) Convert 4B816 to decimal.

Answer: 4B816 = 120810

- *1 mark for place values 16^2, 16^1 and 16^0.*
- *1 mark for correct digit values.*
- *1 mark for correct products.*
- *1 mark for correct addition.*
- *1 mark for the final decimal value.*

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PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 5 and Table 1
z-A+c2/B*D-3A
Figure 5
Write an appropriate program using conect assembly language notations to compute the equation shown in Figure 5 using instructions for zero-address, one-address, and three-address. The available instructions are provided in Table 1. The registers labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register z.
Table 1: lt/achine instruction
Three-address
Zero-address
One-address
ADD (p,q,r)
LOAD p
PUSH p
SUB (p,q,r)
STOR p
POP p
MUL (p,q,r)
ADD p
ADD
DIV (p,q,r)
SUB p
SUB
MUL p
MUL
DIV p
DIV

Source: CSC429 (21).pdf, page(s) 6

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

END OF MARKING SCHEME

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